Network Function Virtualization (NFV)

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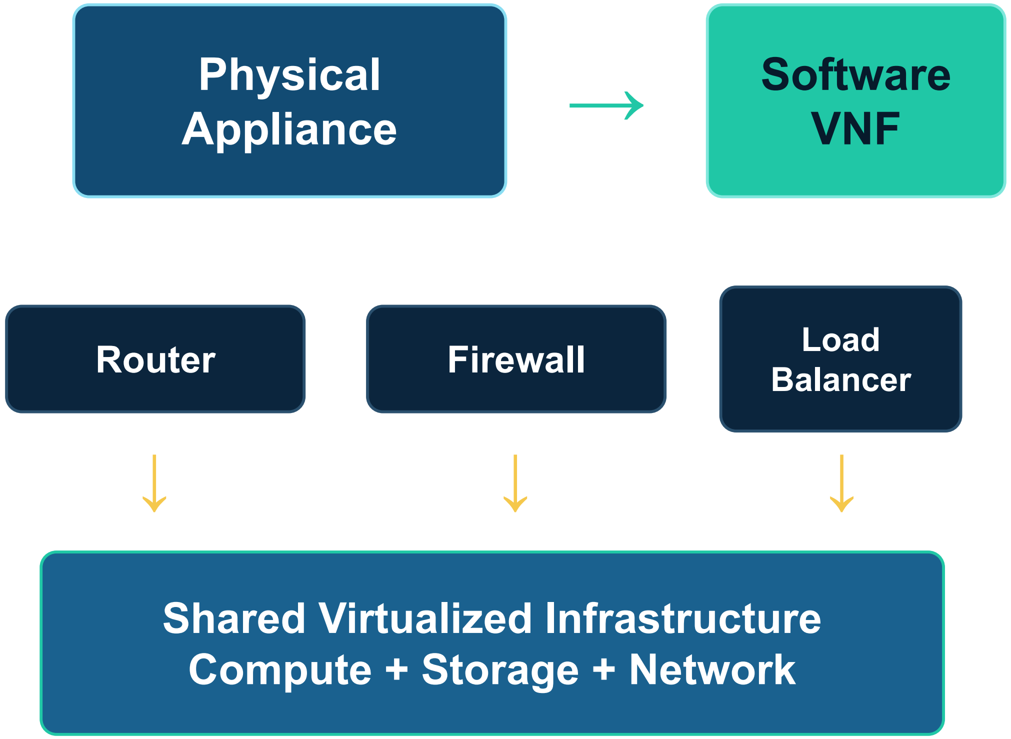
Network Function Virtualization (NFV)

Replacing dedicated routers, firewalls and load balancers with software-based network functions

OSI ARCHITECTURE | VIRTUAL FIREWALL | PERFORMANCE ANALYSIS



Key message: NFV virtualizes the network function, not the OSI model itself.

NFV architecture: the three major blocks

NFV Management & Orchestration (NFV-MANO)

NFVO | VNFM | VIM

Deploys, scales, monitors and terminates VNFs

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Virtual Network Functions (VNFs)

vRouter | vFirewall | vLoad Balancer

↓

NFV Infrastructure (NFVI)

Compute | Storage | Network

Meaning of each term

- NFVI: the physical + virtual resource pool
- VNF: the software network function
- NFV-MANO: lifecycle and resource management
- A service chain can combine multiple VNFs

Example service chain
Client → vFirewall → vLoad Balancer → Server

Experiment design: virtual firewall VNF

Baseline path

Client → **Server**

Measure direct latency, throughput and packet loss.

NFV test path

Client → **vFirewall VNF** → **Server**

Allow TCP 80, block TCP 8080, repeat each test 10 times.

Tools used

- Docker / VirtualBox
- Linux iptables or nftables
- ping for latency / packet loss
- iperf3 for throughput
- Wireshark / tcpdump for packet evidence

Controlled variables
Same machine, packet size, test time, CPU/RAM allocation and network path.

Results and analysis: prove both function and performance

Sample results table

Metric	Baseline	Via vFirewall	Meaning
Avg latency	2.1 ms	2.4 ms	+14.3% overhead
Throughput	92 Mbps	87 Mbps	5.4% lower
Packet loss	0%	0%	Reliable path
TCP 80	Allowed	Allowed	Valid traffic passes
TCP 8080	Allowed	Blocked	Firewall policy works

How to analyze

1. Functional validation
Port 80 passes, Port 8080 is blocked

2. Latency overhead
$(2.4 - 2.1) / 2.1 \times 100 = 14.3\%$

3. Throughput drop
$(92 - 87) / 92 \times 100 = 5.4\%$

Conclusion from sample
NFV adds small overhead, but policy enforcement succeeds.

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